



An Attention-Aware Socratic Pedagogical Intervention System

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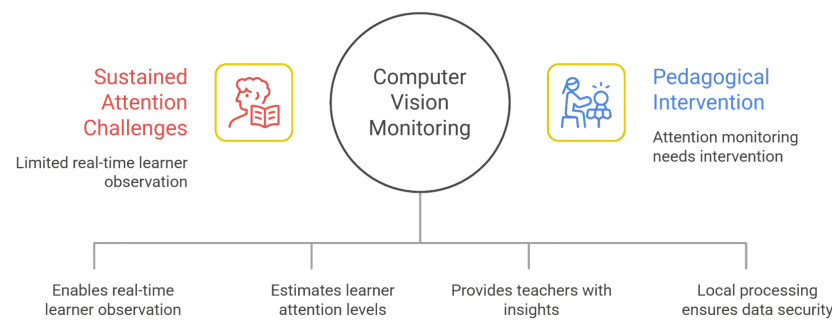
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Abstract

Visoria is an attention monitoring and pedagogical intervention system for online learning. It uses computer vision and machine learning to estimate learner attention from webcam-derived features such as gaze, head pose, face presence, hand activity. Teachers can monitor real-time attention through a web-based dashboard. When sustained low attention is detected, a Socratic intervention engages learners through reasoning, peer comparison, reflection, and reassessment, creating a closed loop: Detect → Intervene → Reassess.

Introduction

Enhancing Online Learning with Computer Vision



Problem Statement

- Attention Decline:** It is difficult to identify when learners gradually lose attention during a learning session.
- Delayed Response:** Learners are often not helped at the right time when they lose attention.
- Passive Learning:** Learners may stay inactive or disengaged without receiving any support or intervention.
- Pedagogical Gap:** Current systems often lack attention-aware strategies to support and re-engage learners.

Objective

- Track learners' attention in real time.
- Estimate attention levels using machine learning.
- Identify when attention stays low for a period of time.
- Start Socratic learning activities when attention drops.
- Encourage learners to think, discuss, and learn with peers.
- Check whether attention improves after the intervention.

System Architecture



Key Feature

- Combines attention monitoring with an actual learning intervention.
- Uses computer vision to notice when a student stays inattentive for a longer period.
- Uses the detected attention state to decide when an intervention may be helpful.
- Gives the student a short activity: Think → Compare → Reflect → Reassess.
- Checks the student's attention again after the activity to see how they respond.

Technology Stack

CATAGORY	TECNOLOGY	
Frontend	REACT JS	TKiner
Backend	node NODE JS	FASTAPI EXPRESS
Database	MONGO DB	
Computer Vision	OpenCV	MEDIAPIPE

References

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Work Flow

